

Algebra II with Trigonometry

Chapter 5.1

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Based on the concepts from your math PDF (angle measure, degrees/radians, coterminal angles, arc length, sector area, circular motion), here are **10 mental math questions**. These can be solved without paper/pencil if you know the key facts from the material:

1. Degrees & Radians

Q1: How many degrees are in π radians?

Q2: What is the radian measure of a right angle?

2. Degree $\leftrightarrow$ Radian Conversions

Q3: Convert 180° to radians.

Q4: What is 1 radian approximately in degrees (rounded to the nearest whole number)?

3. Coterminal Angles

Q5: Name a positive angle that is coterminal with -45° .

Q6: Name a negative angle that is coterminal with 30° .

4. Arc Length ($s = r\theta$)

Q7: A circle has radius 1. What is the length of the arc subtended by an angle of $\pi/2$ radians?

5. Sector Area ($A = \frac{1}{2} r^2\theta$)

Q8: What fraction of the area of a circle is a sector with a central angle of π radians?

6. Circular Motion

Q9: If a point on a circle travels 2π radians in 1 second, how many full revolutions does it make in 1 second?

Q10: If a wheel turns 360° , what is the measure in radians?

Let me know if you'd like answers, explanations, or further questions!